

Virtual Reality-based Digital Shadow of Quadruped Locomotive System

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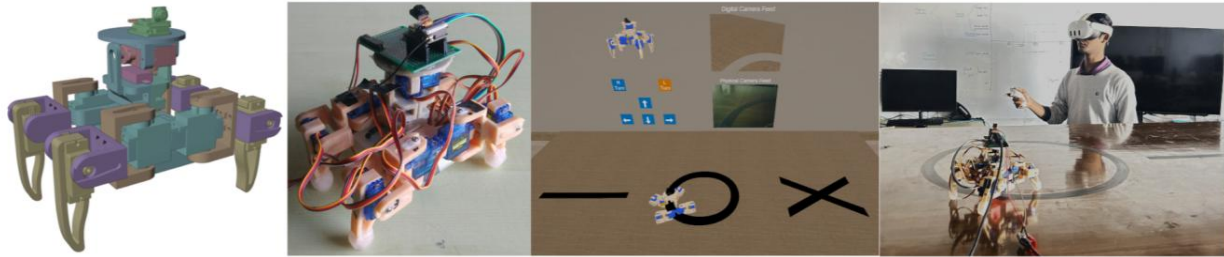


Figure 1: Physical quadruped robot and its virtual model within the VR-based digital shadow environment, demonstrating synchronization between the physical robot and the digital shadow system during teleoperation.

ABSTRACT

Quadruped robots over years have gained significant application due to their advance terrain adaptability and tenet stability compared to wheeled platforms, nonetheless, the locomotion remains one of the most challenging problems in legged robotics over multi-joint coordination, operating in unpredictable environments, and dynamic stability constraints. We propose a low cost open-source quadruped designed to track the head orientation of an operator wearing a virtual reality (VR) headset, while also utilizing joystick commands for locomotion control with the integration of digital shadow to achieve stable locomotion by synchronized digital models of the physical entity for teleoperation applications.

Index terms: Virtual Reality, Locomotion, Navigation, Redirection, Digital Shadow, Immersive Visualisation, Simulation.

1 INTRODUCTION

Quadruped robots have distinct advantages including search-and-rescue operations, industrial inspection, hazardous-environment exploration in navigating challenging terrains, making them highly suitable for operating in uneven and irregular environments. Unlike bipedal systems, quadruped distributes weight across entire body, enabling robust locomotion over uneven surfaces while maintaining balance under external disturbances [1]. Prior quadruped platforms such as Boston Dynamics' Spot [2] and MIT Cheetah [3] demonstrate exceptional dynamic performance but rely on high-cost actuators, proprietary controllers, and computationally intensive architectures.

A quadruped robot has a complex composition, and its stable operation is usually affected by many factors, such as hardware and software system. They achieve efficient locomotion through coordinated gaits, and dynamic weight distribution. Efficient locomotion relies on coordinated gait patterns, body trajectory planning, and stability margins, which are difficult to evaluate

directly on hardware alone [4]. To eliminate this challenge, we propose digital shadow of quadruped locomotive system, where VR-based digital shadow operates in a unidirectional manner to send physical state information and projecting it into simulation for validation, visualization, and gait analysis before optimized commands are executed on hardware.

The system integrates kinematic modelling and simulation-based validation to achieve stable and predictable locomotion by synchronized virtual models of the physical entity.

2 DESIGN & IMPLEMENTATION

The system consists of two components: The physical quadruped robot, responsible for real-world motion execution. The VR-based digital shadow, responsible for visualization. Thus, this work presents a VR-based digital shadow framework for real-time teleoperation and validation of a quadruped locomotion system.

2.1 Physical Quadruped Robot

The physical platform consists of a lightweight, fully 3D-printed quadruped consists of four legs, each with 2 degrees of freedom (DoF), resulting in a total of 8 actuated DoFs for locomotion equipped with ten SG90 servo motors for actuation. Each leg consists of two parts: Hip and Knee joint. In the current setup, eight servos are dedicated to control hip and knee joints movement, while two additional servos enable 2 DoF pan-tilt motion for camera orientation as in Figure 2.

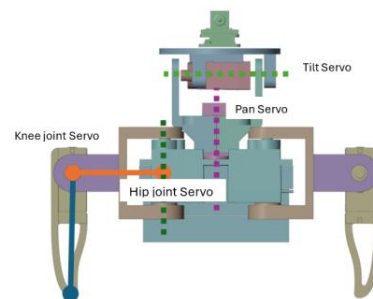


Figure 2: Physical quadruped servo placement and axis of rotation.

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Servo control is handled through a 16-Channel 12-bit PWM servo driver (I2C interface PCA9685), interfaced with an Sseed Studio XIAO ESP32S3 sense microcontroller responsible for command execution, and wireless communication. All actuation and control are performed over Wi-Fi, allowing untethered operation. Power to the ESP32 is provided through an external source, and the servo motors are powered using an AIQU SPS C3010S with the DC power supply of 5 V, 1.3 A, connected to the servo driver driver. Locomotion is controlled by assigning degree-based angles to each joint. These angles are validated in the digital shadow to stay within servo limits and then sent over Wi-Fi as a 1×10 joint-angle matrix, where each value directly assigns the control command to the servo for the physical quadruped movement.

2.2 VR-based Digital Shadow

2.2.1 VR setup

The VR-based high fidelity digital shadow is developed in Unity 6, enabling synchronized simulation, control, and execution of locomotion behaviors of the robot. The digital shadow of the quadruped runs in real time within Unity 6, where joint states, gaits, and trajectories are continuously synchronized with the physical robot. The locomotion is performed through user input from a VR headset (Meta Quest 3) joystick control. The digital shadow observes the locomotion behavior, and validates joint angle commands are transmitted to the robot, ensuring stable motions are executed.

2.2.2 User Interface

The user interface, shown in Figure 3 provides simultaneous visualization of the virtual quadruped, control inputs, and both virtual and physical camera feeds. Using the Meta Quest 3 right controller buttons and joystick control, the user can give directional and turning commands, which are reflected in real time on the 3D model in VR and the physical robot. Furthermore, on-screen four directional and two turning controls are visible, which are forward, backward, left, right, left turn, right turn for visual cues as in Figure 3. Additionally, the ESP32-mounted camera feed and the virtual camera view are displayed concurrently, enabling direct comparison between simulated and real-world behavior.

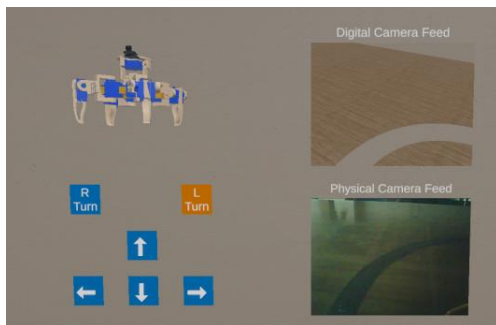


Figure 3: User interface in the VR environment.

2.3 Experiment

The quadruped is operated within a clearly defined workspace for locomotion evaluation, marked with linear (I), circular (O), and sharp turning (X) paths to evaluate the translational motion, and turning behavior of the quadruped for both physical and digital robot as in Figure 4.



Figure 4: Experimental workspace used for locomotion evaluation.

3 RESULTS & DISCUSSION

The physical validation experiments are conducted in a controlled environment, as shown in the Figure 1 and Figure 4. Minor pose discrepancies are clearly observable within the VR environment, and physical quadruped, and are identified through direct visual comparison of synchronized virtual and physical camera feeds and by comparing commanded joint-angle values with the resulting physical pose, thereby demonstrating the effectiveness of the digital shadow in identifying latency, and control disparities.

The presented system establishes a strong foundation for a low cost open-source quadruped robot design and implementation to track the head movements of an operator wearing a virtual reality (VR) headset and establishing digital shadow. To evaluate synchronization accuracy, predefined motion patterns are executed, revealing slight offsets between the physical and virtual models.

4 CONCLUSION AND FUTURE WORK

Through this VR-enabled digital shadow framework, the user can wirelessly control the physical quadruped, observe real-time kinematic changes, and verify synchronized behavior between the digital and physical systems through camera feeds. This design encourages low cost open-source control for a quadruped during teleoperation. In the current setup the battery is not included, and the 10 servos along with the ESP32 board are powered externally. Future work will focus on extending the platform with IMU-based terrain adaptation, integrated power supply for motors and a fully bidirectional digital twin system, along with comprehensive terrain evaluation using Self localization and Mapping Technique.

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